

Data Preconditioned Stochastic Gradients: A new view on data normalization

The preconditioned perpetrators

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Abstract

We devise a data dependent preconditioner for applying gradient descent to linear layers.

Robert: TODO:

- Compare against the implicit preconditioner used by spectral gradient descent, that is $(\mathbf{G}_t \mathbf{G}_t^\top)^{-1/2} \mathbf{G}_t$

1 Introduction

Consider any neural network that has a linear layer. That is, we consider a family of functions

$$h(\mathbf{W}) = h_1(\mathbf{W} h_0(\mathbf{X}_0)),$$

where $\mathbf{X}_0 \in \mathbb{R}^{d_0 \times b}$, b is the batch dimension, and $\mathbf{W} \in \mathbb{R}^{d_2 \times d_1}$ is a matrix of learnable parameters that parametrizes a linear layer. Here h_1 and h_0 could be simple activations, or be themselves a sequences of compositions with learnable parameters. But here we focus only on the $\mathbf{W}$ parameters, since we are interested in how best to update the parameters of a single linear layer. To simplify notation further we also use $\mathbf{X} = h_0(\mathbf{X}_0) \in \mathbb{R}^{d_1 \times b}$ to denote the intermediary embedding, and with an additional loss function ℓ we denote the loss with respect to this linear layer as

$$f(\mathbf{W} \mathbf{X}) := \ell \circ h_1(\mathbf{W} \mathbf{X}). \quad (1)$$

Note that $\mathbf{X}$ depends on the current batch of data, where the training problem with respect to the $\mathbf{W}$ parameters can be represented as

$$\min_{\mathbf{W} \in \mathbb{R}^{d_2 \times d_1}} \mathbb{E}_{\mathbf{X} \sim \mathcal{P}} [f(\mathbf{W} \mathbf{X})], \quad (2)$$

where $\mathbf{X}$ is drawn from a distribution $\mathcal{P}$ that will depend on both the data distribution, and the prior layers of the neural network. For the remainder of this paper will use the notation given in (2), and $\mathbf{X}$ can be interpreted as a batch of data. SGD (Stochastic gradient) type methods applied to (2) are given by

$$\text{Sample } \mathbf{X}_t \sim \mathcal{P} \quad (3)$$

$$\text{Update } \mathbf{W}^{t+1} = \mathbf{W}^t - \gamma \mathbf{G}^t, \quad (4)$$

where $\mathbf{G}^t = \nabla_{\mathbf{W}} f(\mathbf{W} \mathbf{X}_t)|_{\mathbf{W}=\mathbf{W}^t}$ is a stochastic gradient, and γ is the step size.

As our notation (2) highlights, at the t -th step of SGD the current loss over the batch $f(\mathbf{W} \mathbf{X}_t)$ only depends on $\mathbf{W}$ to the extent that it changes the product $\mathbf{W} \mathbf{X}_t$. Using this observation, we derive a data dependent norm for gradient descent, that results in a data preconditioned gradient.

Algorithm 1: DAP: Data Preconditioned gradient method.

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1 Initialize:  $\gamma_t \in [0, 1]$ 
2 for  $t = 0$  to  $T - 1$  do
3   Sample batch  $\mathbf{X}_0 \sim \mathcal{P}$ 
4   Store embedding  $\mathbf{X}_t$  in forward pass
5   Compute gradient  $\mathbf{G}^t = \nabla_{\mathbf{W}} f(\mathbf{W}^t)$ 
6    $\mathbf{P}_t = \text{Schulz}(\mathbf{X}_t \mathbf{X}_t^\top)$ 
7    $\mathbf{W}^{t+1} = \mathbf{W}^t - \gamma_t \mathbf{G}^t \mathbf{P}_t$ 

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Output: $\mathbf{W}^T$

2 A data preconditioned gradient

For the current batch of data $\mathbf{X}_t \sim \mathcal{P}$, where $\mathbf{X}_t \in \mathbb{R}^{d_1, d_0}$, for a step of SGD can be written as

$$\mathbf{W}^{t+1} = \underset{\mathbf{W}}{\operatorname{argmin}} \langle \mathbf{W} - \mathbf{W}^t, \mathbf{G}^t \rangle + \frac{1}{2\gamma} \|\mathbf{W} - \mathbf{W}^t\|^2. \quad (5)$$

where we use $\mathbf{G}^t$ to denote the gradient of this layer. To choose an appropriate norm in (5), we need to consider how changes in $\mathbf{W}$ affect the loss. In particular, changes in $\mathbf{W}$ only matter insofar as they change $\mathbf{W} \mathbf{X}_t$. Consequently we propose using the pseudo-norm

$$\mathbf{W}^{t+1} = \underset{\mathbf{W}}{\operatorname{argmin}} \langle \mathbf{W} - \mathbf{W}^t, \mathbf{G}^t \rangle + \frac{1}{2\gamma} \|(\mathbf{W} - \mathbf{W}^t) \mathbf{X}_t\|^2, \quad (6)$$

where if there is no unique solution to the above, we suppose the least-norm solution is taken. At first glance, the above may have no solution, since it may happen that we can choose $(\mathbf{W} - \mathbf{W}^t) \mathbf{X}_t = 0$ and yet send the linear term $\langle \mathbf{W} - \mathbf{W}^t, \mathbf{G}^t \rangle$ to negative infinity. But this turns out to now be possible because of the properties of the gradient matrix. By the chain-rule we have that

$$\nabla_{\mathbf{W}} f(\mathbf{W} \mathbf{X}) = \nabla_{\mathbf{Z}} f(\mathbf{Z})|_{\mathbf{Z}=\mathbf{W} \mathbf{X}} \mathbf{X}^\top. \quad (7)$$

For short-hand we will use Let $\hat{\mathbf{G}}^t := \nabla_{\mathbf{Z}} f(\mathbf{Z})|_{\mathbf{Z}=\mathbf{W}^t \mathbf{X}_t}$ and $\mathbf{G}^t = \nabla_{\mathbf{W}} f(\mathbf{W}^t \mathbf{X}_t) = \hat{\mathbf{G}}^t \mathbf{X}_t^\top$, thus

$$\mathbf{G}^t = \hat{\mathbf{G}}^t \mathbf{X}_t^\top. \quad (8)$$

Using the above identity in (6) gives

$$\mathbf{W}^{t+1} = \underset{\mathbf{W}}{\operatorname{argmin}} \left\langle (\mathbf{W} - \mathbf{W}^t) \mathbf{X}_t, \hat{\mathbf{G}}^t \right\rangle + \frac{1}{2\gamma} \|(\mathbf{W} - \mathbf{W}^t) \mathbf{X}_t\|^2, \quad (9)$$

This problem now always has a solution, and a unique least-norm solution.

Lemma 2.1. For $\mathbf{W}^t, \mathbf{G}^t \in \mathbb{R}^{d_2 \times d_1}$ and $\mathbf{X}_t \in \mathbb{R}^{d_1 \times d_0}$, the least-norm solution with respect to $\|\mathbf{W} - \mathbf{W}^t\|_F$ of

$$\mathbf{W}^{t+1} = \underset{\mathbf{W}}{\operatorname{argmin}} \left\langle \mathbf{W} - \mathbf{W}^t, \mathbf{G}^t \right\rangle + \frac{1}{2\gamma} \|(\mathbf{W} - \mathbf{W}^t) \mathbf{X}_t\|_F^2, \quad (10)$$

is given by

$$\mathbf{W}^{t+1} = \mathbf{W}^t - \gamma \mathbf{G}^t (\mathbf{X}_t \mathbf{X}_t^\top)^\dagger. \quad (11)$$

Proof. For brevity let $\Delta \mathbf{W} := \mathbf{W} - \mathbf{W}^t$, $\mathbf{X} = \mathbf{X}_t$ and $\mathbf{G} = \mathbf{G}^t$.

Considering the equivalent problem (9), and completing the squares gives

$$\min_{\Delta \mathbf{W}} \|\Delta \mathbf{W} \mathbf{X} + \gamma \hat{\mathbf{G}}\|_F^2. \quad (12)$$

The least norm solution with respect to $\|\Delta \mathbf{W}\|$ is now given by taking the pseudo-inverse of $\mathbf{X}$, gives

$$\Delta \mathbf{W} = -\gamma \hat{\mathbf{G}} \mathbf{X}^\dagger.$$

Now using the identity

$$\mathbf{X}^\dagger = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^\dagger$$

gives

$$\Delta \mathbf{W} = -\gamma \hat{\mathbf{G}} \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^\dagger = -\gamma \mathbf{G} (\mathbf{X} \mathbf{X}^\top)^\dagger,$$

which is the solution. $\square$

Lemma 2.2. Let $\|\cdot\|$ be a unitarily invariant norm. Consider an arbitrary real matrix $\mathbf{G}$. Define

$$\mathbf{G}_* := \operatorname{argmax}_{\|\mathbf{T}\|=1} \langle \mathbf{T}, \mathbf{G} \rangle.$$

David: Uniqueness??? If $\mathbf{V}$ is a matrix with orthonormal columns, then

$$\mathbf{G}_* \mathbf{V}^\top = (\mathbf{G} \mathbf{V}^\top)_*$$

Proof. **David:** This might be in a textbook Note that by Hölder's inequality [bhatia] we have $\langle \mathbf{T}, \mathbf{G} \mathbf{V}^\top \rangle \leq \|\mathbf{T}\| \|\mathbf{G} \mathbf{V}^\top\|_* = \|\mathbf{G}\|_*$, where we used that $\|\cdot\|_*$ is unitarily invariant [bhatia]. Equality can be achieved by setting $\mathbf{T} = \mathbf{G}_* \mathbf{V}^\top$. $\square$

Lemma 2.3. Let $\mathbf{W}^t, \mathbf{G}^t \in \mathbb{R}^{d_2 \times d_1}$ and $\mathbf{X}_t \in \mathbb{R}^{d_1 \times d_0}$. For any unitarily invariant norm $\|\cdot\|$, the solution [David: uniqueness?](#) to

$$\min_{\mathbf{W}} \langle \mathbf{W} - \mathbf{W}^t, \mathbf{G}^t \rangle + \frac{1}{2\gamma} \|(\mathbf{W} - \mathbf{W}^t) \mathbf{X}_t\|^2, \quad (13)$$

is given by

$$\mathbf{W}^{t+1} = \mathbf{W}^t - \gamma \|\mathbf{G}^t \mathbf{X}_t^{\dagger \top}\|_* (\mathbf{G}^t \mathbf{X}_t^{\dagger \top})_* \mathbf{X}_t^\dagger \quad (14)$$

where thus star operator returns the dual gradient, that is [David: Uniqueness???](#)

$$\mathbf{G}_* := \operatorname{argmax}_{\|\mathbf{T}\|=1} \langle \mathbf{T}, \mathbf{G} \rangle \quad (15)$$

$$\|\mathbf{G}\|_* = \max_{\|\mathbf{T}\|=1} \langle \mathbf{T}, \mathbf{G} \rangle \quad (16)$$

Proof. Define $\Delta \mathbf{W}^t := \mathbf{W} - \mathbf{W}^t$. By recalling $\mathbf{G}_t = \hat{\mathbf{G}}_t \mathbf{X}_t^\top$ we can consider the equivalent problem (9)

$$\min_{\Delta \mathbf{W}^t} \langle \Delta \mathbf{W}^t \mathbf{X}_t, \hat{\mathbf{G}}^t \rangle + \frac{1}{2\gamma} \|\Delta \mathbf{W}^t \mathbf{X}_t\|^2 \quad (17)$$

Let $\mathbf{P}_{\mathbf{X}_t}$ denote the orthogonal projection onto $\operatorname{range}(\mathbf{X}_t)$. Note that if $\Delta \mathbf{W}^t$ is a solution to (17) then so is $\Delta \mathbf{W}^t \mathbf{P}_{\mathbf{X}_t}$. We will constrain our solution set to be least $\|\cdot\|$ -norm solutions. By Fan's dominance theorem [bhatia] we have $\|\Delta \mathbf{W}^t \mathbf{P}_{\mathbf{X}_t}\| \leq \|\Delta \mathbf{W}^t\|$. Hence, a least $\|\cdot\|$ -norm solution satisfies

$$\Delta \mathbf{W}^t = \Delta \mathbf{W}^t \mathbf{P}_{\mathbf{X}_t} \Leftrightarrow \operatorname{rowspace}(\Delta \mathbf{W}^t) \subset \operatorname{range}(\mathbf{X}_t).$$

In other words, if $\mathbf{X}_t = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$ is the rank-reduced SVD of $\mathbf{X}_t$ so that $\mathbf{\Sigma}$ is a square full-rank diagonal matrix, we have

$$\Delta \mathbf{W}^t = \mathbf{K} \mathbf{U}^\top,$$

for some matrix $\mathbf{K}$. Substituting $\Delta \mathbf{W}^t \mathbf{X}_t = \mathbf{K} \mathbf{\Sigma} \mathbf{V}^\top$ into (17) and using that $\|\cdot\|$ is unitarily invariant yields the following reformulation of (17)

$$\min_{\mathbf{K}} \langle \mathbf{K} \mathbf{\Sigma}, \hat{\mathbf{G}}^t \mathbf{V} \rangle + \frac{1}{2\gamma} \|\mathbf{K} \mathbf{\Sigma}\|^2.$$

Noting that $\mathbf{\Sigma}$ is square non-singular and substituting $\hat{\mathbf{K}} = \mathbf{K} \mathbf{\Sigma}$ yields

$$\min_{\hat{\mathbf{K}}} \langle \hat{\mathbf{K}}, \hat{\mathbf{G}}^t \mathbf{V} \rangle + \frac{1}{2\gamma} \|\hat{\mathbf{K}}\|^2.$$

Note that since $\Delta \mathbf{W}^t = \hat{\mathbf{K}} \mathbf{\Sigma}^{-1} \mathbf{U}^\top$, $\mathbf{\Sigma}$ is non-singular, and $\mathbf{U}$ has orthogonal columns, we can conclude $\Delta \mathbf{W}^t = \mathbf{0}$ if and only if $\hat{\mathbf{K}} = \mathbf{0}$. Hence, we can always write $\hat{\mathbf{K}} = c \mathbf{T}$ where $c \geq 0$ and $\mathbf{T}$ is some matrix satisfying $\|\mathbf{T}\| = 1$. Inserting this formulation gives the the following optimization problem over $c \geq 0$ and $\mathbf{T}$

$$\min_{\|\mathbf{T}\|=1, c \geq 0} c \langle \mathbf{T}, \hat{\mathbf{G}}^t \mathbf{V} \rangle + \frac{c^2}{2\gamma} = \min_{c \geq 0} c \left(\min_{\|\mathbf{T}\|=1} \langle \mathbf{T}, \hat{\mathbf{G}}^t \mathbf{V} \rangle \right) + \frac{c^2}{2\gamma}.$$

Hence, $\mathbf{T} = -(\hat{\mathbf{G}}^t \mathbf{V})_*$ and $c = \gamma \|\hat{\mathbf{G}}^t \mathbf{V}\|_*$. Consequently,

$$\Delta \mathbf{W}^t = c \mathbf{T} \Sigma^{-1} \mathbf{U}^\top = -\gamma \|\hat{\mathbf{G}}^t \mathbf{V}\|_* (\hat{\mathbf{G}}^t \mathbf{V})_* \Sigma^{-1} \mathbf{U}^\top.$$

Noting that $\mathbf{G}^t \mathbf{X}_t^{\dagger\top} = \hat{\mathbf{G}}^t \mathbf{X}_t^\top \mathbf{X}_t^{\dagger\top} = \hat{\mathbf{G}}^t \mathbf{V} \mathbf{V}^\top$, using [Lemma 2.2](#), and that the dual norm of a unitarily invariant norm is also unitarily invariant [[bhatia](#)] gives

$$\begin{aligned} \Delta \mathbf{W}^t &= -\gamma \|\hat{\mathbf{G}}^t \mathbf{V}\|_* (\hat{\mathbf{G}}^t \mathbf{V})_* \Sigma^{-1} \mathbf{U}^\top \\ &= -\gamma \|\hat{\mathbf{G}}^t \mathbf{V} \mathbf{V}^\top\|_* (\hat{\mathbf{G}}^t \mathbf{V})_* \mathbf{V}^\top \mathbf{V} \Sigma^{-1} \mathbf{U}^\top \\ &= -\gamma \|\hat{\mathbf{G}}^t \mathbf{V} \mathbf{V}^\top\|_* (\hat{\mathbf{G}}^t \mathbf{V} \mathbf{V}^\top)_* \mathbf{V} \Sigma^{-1} \mathbf{U}^\top \\ &= -\gamma \|\mathbf{G}^t \mathbf{X}_t^{\dagger\top}\|_* (\mathbf{G}^t \mathbf{X}_t^{\dagger\top})_* \mathbf{X}_t^\dagger. \end{aligned}$$

Recalling that $\Delta \mathbf{W}^t = \mathbf{W} - \mathbf{W}^t$ yields the desired result. $\square$

David: [Old stuff commented here](#)

Nikhil: Note that if we define $\mathbf{C} := \mathbf{X} \mathbf{X}^\top$, then

$$\begin{aligned} \|\mathbf{W} \mathbf{X}\|_{\text{op}}^2 &= \lambda_{\max}(\mathbf{W} \mathbf{X} \mathbf{X}^\top \mathbf{W}) \\ &= \lambda_{\max}(\mathbf{W} \mathbf{C} \mathbf{W}) \\ &= \|\mathbf{W} \mathbf{C}^{1/2}\|_{\text{op}}^2. \end{aligned}$$

Therefore in the above we can replace $\mathbf{X}_t^\dagger$ with $\mathbf{C}^{-1/2}$. The upshot is that now $\Delta \mathbf{W}^t$ is efficiently computable with no batch size dependence. Equivalently we can think of the regularizer being given by the $\|\cdot\|_{\mathbf{C}^{-1} \rightarrow \ell_2}$ operator norm. In KFAC style we can also use the backward covariance to change the output space norm.